Compute Marketplace - Job Lifecycle

Compute Buyer
Compute Provider
Frontend
Server (Routes)
Match Manager
Blockchain Manager
Provider-Client Manager
MongoDB
Buyer Docker Registry
Third-Party Object Storage
Provider Client App
Blockchain / Smart Contract

Group [Provider Onboarding]
1: Log in, link wallet
2: Link wallet request
3: Authorize wallet
4: Create provider account document
5: Saved
6: Account linked

Group [Buyer Onboarding]
7: Log in, link wallet
8: Link wallet request
9: Authorize wallet
10: Create buyer account document
11: Saved
12: Account linked

Group [Provider Benchmark]
13: Install client, run benchmark
14: Submit benchmark results
15: Update provider hardware score
16: Saved
17: Ack

Group [Job Submission and Matching]
18: Submit job
19: Submit job request
20: Store job
21: Wake up
22: Scan for matching providers
23: Matching providers
24: Call startJob route
25: startJob for selected providers
26: startJob route

Group [Accept Job]
27: Pull buyer container image
28: Image
29: acceptJob (session key, stake)
30: Registered as provider for job

Group [Commit Phase (all providers)]
31: Run job, write to standardized output dir
32: Hash output
33: Encrypt hash
34: commitResult(encrypted hash)

Group [Reveal Phase (all providers, no buyer involvement)]
35: Post decryption key
36: Decrypt hashes, verify, perform consensus
37: Consensus event emitted

Group [Result Retrieval and Payment]
38: Provider selected in consensus
39: Request output
40: getOutput route
41: Output
42: Output
43: Validate output hash
44: Sign verifyReceipt
45: verifyReceipt
46: Pay transfer
47: Store computation result
48: Stored
49: Request results
50: Fetch results
51: Results
52: Results

Compute Buyer
Compute Provider
Frontend
Server (Routes)
Match Manager
Blockchain Manager
Provider-Client Manager
MongoDB
Buyer Docker Registry
Third-Party Object Storage
Provider Client App
Blockchain / Smart Contract